

PROJECT DESCRIPTION

Sindh Solar Energy Project (SSEP) — 500 MW PV, Pakistan

Format: Structured Q&A for Political Risk Underwriting

This document responds to the standard underwriter question set used by the Lloyd's political risk market for project finance transactions. Each question is answered by Standard Chartered Bank's Project Finance team in collaboration with Aon Specialty. This format has been adopted to directly address underwriter information requirements without the need for supplementary clarification requests.

SECTION A: PROJECT IDENTITY & STRUCTURE

	What is the project and where is it located?
Q1	The Sindh Solar Energy Project (SSEP) is a 500 MW AC ground-mounted photovoltaic solar power plant located in the Gharo Solar Energy Zone, District Thatta, Sindh Province, Pakistan (GPS: 24.7°N, 67.6°E). It is one of the largest utility-scale solar projects in Pakistan's history and forms part of the government's Indicative Generation Capacity Expansion Plan (IGCEP) under the National Energy Policy 2023-2027. The project site covers approximately 2,200 acres of government-leased land. Risk Relevance: <i>Material to: Asset Valuation (CEND) / Geographic Risk Assessment (Political Violence)</i>

	Who are the project company shareholders and what is the ownership structure?
Q2	The project company, Sindh Solar Energy Project (Private) Limited (SSEP SPV), is incorporated in Pakistan under the Companies Act 2017. Shareholding: Sindh Industrial & Development Authority (SIDA) — 60% (provincial government entity); Envision Energy Pakistan Holdings Ltd — 30% (wholly owned subsidiary of Envision Energy Co., Ltd, China); GreenPower Investments Ltd (BVI) — 10% (private equity co-investor). Standard Chartered Bank holds no equity in the SPV. SCB's relationship is solely as senior secured lender and facility agent. Risk Relevance: <i>Material to: Sovereign nexus assessment (CEND / Non-Payment), State-owned entity involvement</i>

	What is the technology and who is the EPC contractor?
Q3	The project uses bifacial crystalline silicon photovoltaic modules (550W monocrystalline, N-type TOPCon technology) supplied by Longi Green Energy Technology Co., Ltd. (China). The EPC contract is a fixed-price, fixed-date lump-sum turnkey contract awarded to Envision Energy (China) — reference SSEP/EPC/2024/001. Contract value: USD 310,000,000. Completion guaranteed by 30 June 2026. Performance liquidated damages: 10% of contract value. Envision Energy has delivered 3.4 GW of solar EPC in Asia — reference project list provided separately. O&M post-commissioning will be performed by SIDA Renewable Energy Services (Pvt) Ltd, a SIDA subsidiary. Risk Relevance: <i>Material to: Project completion risk (affects onset of insurable interest); EPC contractor credit quality</i>

SECTION B: REVENUE, OFFTAKE & SOVEREIGN EXPOSURE

	What is the revenue model and who is the offtaker?
Q4	Revenue is generated entirely under a 25-year Power Purchase Agreement (PPA) signed 14 August 2024 between SSEP SPV and the National Transmission & Despatch Company (NTDC) — Pakistan's national electricity transmission utility, wholly owned by the Government of Pakistan (GoP). The PPA tariff is USD 5.2 cents/kWh (real, USD-denominated, indexed to US CPI annually from Year 3). NTDC takes 100% of the plant's net generation (no curtailment risk above 95% dispatch obligation). The PPA was approved by the National Electric Power Regulatory Authority (NEPRA) — NEPRA approval reference: NEPRA/R/SSEP/2024-025 dated 3 October 2024. Risk Relevance: <i>Material to: Sovereign Non-Payment / Contract Frustration — NTDC is the sovereign-linked obligor; USD tariff reduces currency risk</i>

	What government support instruments are in place?
Q5	The Government of Pakistan (acting through the Ministry of Finance) has issued: (1) A Sovereign Guarantee covering NTDC's payment obligations under the PPA — reference MOF/SOVGUARANTEE/2024/SSEP, dated 3 September 2024. The guarantee is unconditional, irrevocable, and payable on demand within 45 business days of a valid claim notice. (2) A

	Letter of Support from the Ministry of Energy (Power Division) confirming the project's status as a Priority Infrastructure Project under the IGCEP. (3) A ESIA no-objection certificate from the Pakistan Environmental Protection Agency (Ref: Pak-EPA/ESIA/2024/Sindh/SSEP). Both documents have been reviewed and opined upon by local counsel (Orr, Dignam & Co., Karachi — legal opinion dated 18 November 2024). Risk Relevance: <i>Material to: Sovereign Non-Payment coverage — enforceability and validity of Sovereign Guarantee is the primary risk transfer mechanism; legal opinion is key</i>
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	How are revenues structured to facilitate USD debt service?
Q6	The PPA tariff is denominated in USD and paid by NTDC monthly in arrears via a Revenue Escrow Account maintained at Standard Chartered Bank, Karachi Branch. Funds are swept daily into the offshore USD Debt Service Reserve Account (DSRA) held at SCB London. Waterfall: (1) O&M costs (PKR-denominated portion converted at prevailing SBP rate); (2) Senior Debt Service (USD — paid directly from DSRA); (3) CAPEX reserve; (4) Dividend distribution to shareholders. The DSRA maintains a 6-month debt service forward looking reserve at all times. This structure substantially isolates USD debt service from PKR transfer restriction risk. Risk Relevance: <i>Material to: Currency Inconvertibility — residual exposure is limited to PKR O&M conversion; escrow structure is a key underwriting comfort</i>

SECTION C: COUNTRY & POLITICAL RISK ASSESSMENT

	What is Pakistan's sovereign credit rating and economic outlook?
Q7	Pakistan sovereign ratings as of January 2025: S&P — CCC+ (Stable); Moody's — Caa2 (Positive); Fitch — CCC+. Pakistan concluded a USD 3 billion IMF Stand-By Arrangement in April 2023 and entered a new USD 7 billion Extended Fund Facility (EFF) in September 2024 (approved by IMF Board — reference EFF/PAK/2024/047). FX reserves have recovered to USD 10.8 billion (equivalent of approximately 2.3 months import cover) as at December 2024. GDP growth forecast: 3.2% for FY2025 (IMF World Economic Outlook, October 2024). The active IMF programme significantly reduces near-term sovereign default risk but is conditional on fiscal reform implementation. Risk Relevance: <i>Material to: Sovereign Non-Payment probability assessment; Transfer Restriction risk (FX reserve adequacy)</i>

	What is the history of Pakistan's treatment of IPP investors, specifically in the power sector?
Q8	Pakistan has a chequered history with Independent Power Producers (IPPs). In 2020-2021, the government initiated a review of IPP contracts signed in the 1990s-2000s, alleging overpricing, and secured renegotiated tariffs from several large thermal IPPs under pressure. However, renewable energy projects — particularly those structured post-2018 under the New CPEC/IGCEP framework — have not faced equivalent renegotiation pressure. SSEP falls within the new NEPRA-approved framework which incorporates risk-sharing provisions specifically designed to avoid NTDC non-payment situations (mandatory 45-day payment cure and escalation to MoF under the Sovereign Guarantee). SCB's Project Finance team considers the most material political risk to be payment delays rather than outright contract repudiation. Risk Relevance: <i>Material to: Contract Frustration / Creeping Expropriation — context critical for underwriters assessing probability and severity</i>

	What physical security and political violence risks affect the project site?
Q9	The Gharo Solar Energy Zone in Sindh Province is geographically remote and historically has not been subject to significant terrorist or militant activity compared to Khyber Pakhtunkhwa or Balochistan provinces. The primary risks are: (1) General civil unrest in the context of political instability (Pakistan experienced significant political turmoil in 2022-2023 following PM Khan's ouster); (2) Low-level crime and opportunistic vandalism of infrastructure. A site security assessment was conducted by G4S Risk Consulting (January 2025) — the assessment rated the site as MEDIUM-LOW political violence risk, primarily based on Sindh Province's relative stability. A perimeter security plan has been agreed with SIDA and local law enforcement (Sindh Police). The plant will not employ armed private security; community liaison officers will be deployed. Risk Relevance: <i>Material to: Political Violence peril assessment — probability, asset vulnerability, and maximum foreseeable loss estimation</i>

SECTION D: KEY DOCUMENTS AVAILABLE

Document	Status	Availability
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Power Purchase Agreement (NTDC–SSEP SPV, 25 years)	Executed	On NDA request
GOP Sovereign Guarantee (MOF/SOVGUARANTEE/2024/SSEP)	Original — notarised	On NDA request
NEPRA Approval (NEPRA/R/SSEP/2024-025)	Regulatory approval	Attached
EPC Contract (SSEP/EPC/2024/001 — Envision Energy)	Executed, lump-sum	Executive summary only
IMF EFF Programme — Pakistan (EFF/PAK/2024/047)	Public document	Available
G4S Security Assessment — Gharo Site (Jan 2025)	Final report	On NDA request
Local Legal Opinion — Orr Dignam & Co. (Nov 2024)	Final opinion	On NDA request
NEPRA Generation Licence (SSEP SPV)	Issued 5 Dec 2024	Attached
Financial Model — Base, Upside, Stress Cases	Locked model	On NDA request

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